

Zebra: Missing copy constraint in halo2_gadgets variable-base scalar multiplication allows under-constrained base, breaking Orchard Action circuit soundness

Report Type: Weekly · Generated: July 06, 2026 21:46 UTC · Coverage: Jul 06 - Jul 06, 2026

1 Total Advisories	1 Critical	0 High	0 Medium
CRITICAL Severity	9.3 CVSS / Risk Score	TLP:CLEAR TLP Classification	PRO Customer Tier

Executive Summary

Summary A soundness vulnerability in the variable-base scalar multiplication gadget of `halo2\_gadgets` allowed a malicious prover to produce a valid proof for an Orchard Action with an *under-constrained* base point. Because this gadget enforces the diversified-address-integrity condition of the Orchard Action statement, the flaw let a prover satisfy that condition for an arbitrary (pk_d, g_d, ivk) triple, effectively bypassing the check that binds an Action to the correct incoming viewing key — and therefore to the correct nullifier (nf) and spend validating key (ak) — of the note being spent. The main practical consequence is that an adversary could have performed a *double-spend within the Orchard pool*, resulting in a balance violation: the same note could be spent multiple times, each time revealing a distinct, valid-looking nullifier. The total ZEC supply was protected by Zcash's turnstile mechanism, which bounds value flowing out of any pool, so unbou

CONFIDENTIALITY NOTICE: This report is classified PRO tier and intended exclusively for the named recipient. Redistribution, resale, or disclosure to unauthorized parties is strictly prohibited. © 2026 CYBERDUDEBIVASH SENTINEL APEX. All rights reserved.

Top Threat Advisories

High-priority advisories ranked by CVSS score, exploitability, and actor attribution.

Advisory Title	Severity	CVSS	EPSS	AI Summary
Zebra: Missing copy constraint in halo2_gadgets variable-base scalar	Critical	9.8	-	### Summary A soundness vulnerability in the variable-base scalar multiplication

MITRE ATT&CK® v15 Mapping

Technique density score: 0.0 · Techniques observed: 0 · Tactics covered: 0

MITRE ATT&CK techniques pending analyst enrichment. Deploy detection rules from Section 6 to identify related activity.

Threat Actor Intelligence

Attribution analysis across advisories in this report period.

Threat Actor	Count	Associated CVEs / Indicators
CDB-UNATTR-CVE	1	-

Indicators of Compromise (IOCs)

IOC extraction for this advisory is pending enrichment pipeline completion. PRO subscribers receive real-time IOC push to SIEM/SOAR via the SENTINEL APEX webhook at: **GET /api/v1/intel/iocs?advisory={id}**. Full IOC packages include IPv4, domain, SHA256, and URL indicators with confidence scores, first-seen timestamps, and STIX 2.1 formatted bundles.

Detection Engineering

Deploy these production-ready detection rules into your SIEM platform to identify exploitation attempts in real time. Compatible with Microsoft Sentinel, Splunk ES, and any Sigma-compatible platform.

Sigma Rule - Webserver / Process Monitoring

```
title: SENTINEL APEX - CDB-ADVISORY Exploitation Attempt
id: cdb-cdb-advisory-det
status: experimental
description: Detects exploitation indicators related to CDB-ADVISORY.
references:
- https://intel.cyberdudebivash.com
author: CyberDudeBivash SENTINEL APEX
date: 2026/07/06
tags:
- attack.initial_access
- attack.t1190
logsource:
category: webserver
detection:
keywords:
- 'CDB-ADVISORY'
condition: keywords
falsepositives:
- Security testing / penetration testing
level: medium
```

Microsoft Sentinel - KQL Query

```
// SENTINEL APEX - Zebra: Missing copy constraint in halo2_gadgets variable-base scalar multiplication allows
under-constrained base, breaking Orchard Action circuit soundness
// Advisory | Microsoft Sentinel KQL
let time_window = ago(24h);
SecurityAlert
| where TimeGenerated > time_window
| where Description has "Zebra: Missing copy constraint in halo2_"
or AlertName has "Zebra: Missing copy constraint in halo2_"
or ExtendedProperties has "Zebra: Missing copy constraint in halo2_"
| extend Rule = "APEX-ADVISORY", Severity = "CRITICAL"
| project TimeGenerated, AlertName, AlertSeverity, Entities, Description, Rule, Severity
| order by TimeGenerated desc
```

Incident Response Playbook

Phase 1 - 0-24 Hours (Containment)

1. Activate IR team; assign lead analyst and executive sponsor.
2. Isolate or WAF-shield Zebra: Missing copy constraint in halo2_gadgets variable instances exposed to the internet.
3. Enable verbose access logging on all Zebra: Missing copy constraint in halo2_gadgets variable endpoints immediately.

- 4. Pull last 72h of web server and application logs for forensic baselining.
- 5. Check SENTINEL APEX IOC feeds and SIEM alerts for exploitation hits in your environment.
- 6. Notify relevant internal stakeholders and legal / compliance team.

Phase 2 - 24-72 Hours (Eradication)

- 1. Apply vendor patch for Zebra: Missing copy constraint in halo2_gadgets variable or implement recommended mitigation.
- 2. Deploy Sigma and KQL detection rules from Section 6 across all SIEM platforms.
- 3. Rotate all API keys, service account credentials, and session tokens.
- 4. Conduct forensic timeline reconstruction for potential compromise window.
- 5. Validate patch deployment across 100% of affected instances via asset inventory.

Phase 3 - 7 Days (Recovery & Hardening)

- 1. Conduct post-incident review and update incident response runbooks.
- 2. Threat hunt for lateral movement or persistence artifacts using MITRE ATT&CK mapping.
- 3. Update asset inventory with patched version information and closure evidence.
- 4. Submit findings to internal risk register and CISO dashboard.
- 5. Schedule follow-up vulnerability scan to confirm remediation completeness.

Financial Impact Analysis (FAIR Model)

\$2.4M - \$12M Breach Cost Range (FAIR)	\$5.8M IBM Cost of Breach 2025 Median	\$1.2M/day Business Interruption Cost	\$850K Regulatory Fine Exposure
--	--	--	--

Critical severity breaches trigger mandatory 72-hour regulatory disclosure (GDPR Art. 33), class-action exposure, and long-tail legal costs averaging 18-36 months post-incident.

Remediation Cost Estimate: \$380K · Includes engineering time, IR retainer activation, forensic investigation, and customer notification costs.

Regulatory & Compliance Implications

Organizations processing this advisory must evaluate obligations under applicable regulatory frameworks. The table below maps this threat to relevant compliance requirements.

Framework	Reference	Obligation
GDPR	Art. 33 / Art. 83(4)	72-hour breach notification to DPA; fines up to 4% of global annual turnover
PCI-DSS v4.0	Req. 6.3 / 6.4	Critical patches within 1 month; compensating controls required
NIS2 Directive	Art. 23	Early warning within 24h; full incident report within 72h

Framework	Reference	Obligation
SOX / ICFR	Section 302 / 404	Material cybersecurity weakness disclosure in 10-K/10-Q
ISO 27001:2022	A.8.8 / A.5.29	Documented vulnerability management and incident management

Actionable Recommendations

Prioritized defensive actions based on this period's intelligence.

1. Apply patches immediately for Zebra: Missing copy constraint in halo2_gadgets variable-base scalar multiplication allows under-constrained base, breaking Orchard Action circuit soundness.
2. Enable enhanced monitoring for related IOCs.
3. Review SIEM rules for associated MITRE ATT&CK techniques.
4. Apply vendor patch for Zebra: Missing copy constraint in halo2_gadgets variable-base scalar multiplication allows under-constrained base, breaking Orchard Action circuit soundness immediately - check NVD for official fix guidance.
5. Enable enhanced logging and alerting on all systems running affected software versions.
6. Deploy the Sigma detection rule from this report into your SIEM platform (Sentinel / Splunk / Elastic).
7. Audit all internet-facing instances of affected products and confirm patch deployment.

Appendix - Report Metadata & Legal

Field	Value
Report ID	intel--aa9e5103b58b00ba750ce6b5
Report Type	Weekly
Platform	CYBERDUDEBIVASH SENTINEL APEX v166.0 GOD-MODE
Generated At	2026-07-06T21:46:09.38540
Customer Tier	PRO
Customer Email	-
TLP	TLP:CLEAR
STIX Version	2.1
ATT&CK Version	v15

Legal & Usage Notice

This threat intelligence report is produced by CYBERDUDEBIVASH SENTINEL APEX, operated by Bivash Nath (GSTIN: 21ARKPN8270G1ZP). All intelligence is aggregated from public vulnerability databases (NVD, CVSS), threat feeds, and proprietary AI enrichment pipelines. This report is provided AS-IS for informational and defensive security purposes only. CYBERDUDEBIVASH accepts no liability for actions taken based on this intelligence. Redistribution, resale, or disclosure to unauthorized parties is strictly prohibited without written consent. For licensing inquiries: enterprise@cyberdudebivash.com